

Small-business loan default: a scorecard tested the way a lender would use it

Predicting early charge-off of SBA 7(a) loans approved after the model was built, benchmarked against the lender's own price for risk, with the checks a model-risk team runs.

Andres Perez Python · DuckDB · scikit-learn · R Shiny Code: github.com/Andresperez397/sba-loan-default-risk

Live app: sba-loan-default-risk.share.connect.posit.cloud

DATA

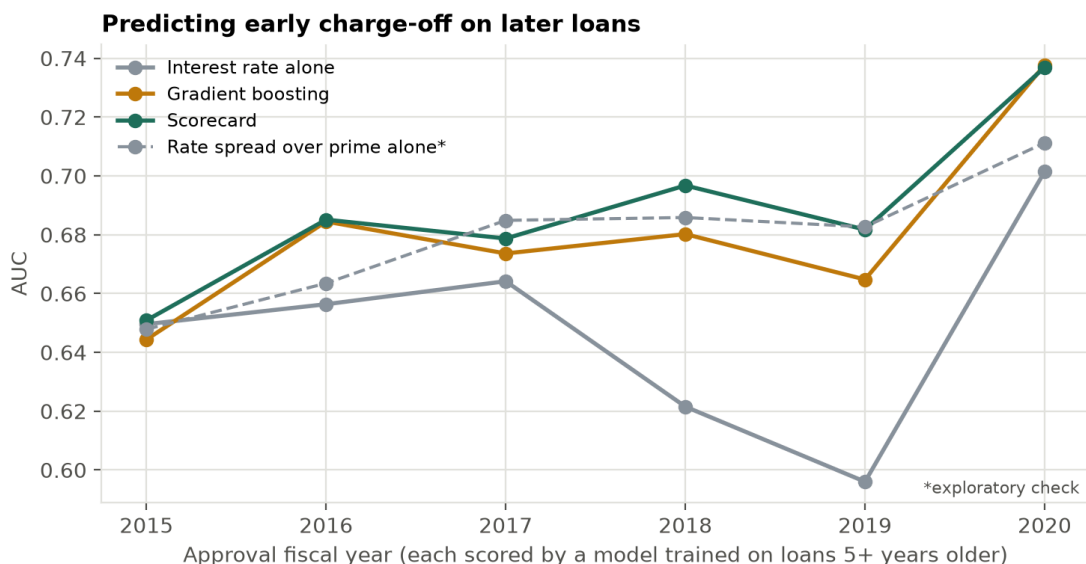
514,984 disbursed SBA 7(a) loans approved FY2010–FY2020 (SBA loan-level FOIA data, public domain). The outcome is a charge-off within five years of approval. Test cohorts FY2015–FY2020 (304,289 loans) are each scored by a model trained only on loans five or more years older, matching what a lender could have known.

APPROACH

- **Audit, then a frozen plan;** borrower names and addresses never read.
- **Leak caught:** SBA overwrites the term of loans that went bad (AUC 0.86 on its own), so term was removed.
- **Models:** weight-of-evidence scorecard (points, reason codes) against gradient boosting and an interest-rate baseline.
- **Policy and model risk:** approval trade-offs in dollars, stability (PSI), calibration and segments.

FINDINGS

- **The scorecard beats boosting out of time** (AUC 0.684 against 0.677), even though boosting won in-time cross-validation.
- **The edge over pricing is modest:** +0.052 AUC against the absolute rate, but most of that was the prime rate rising. Against the rate's spread over prime, +0.010 (exploratory).
- **Policy:** the scorecard cuts the count of early charge-offs at 70–75% approval, but in lender dollars declining by rate is as good or slightly better, because the scorecard declines many small loans.
- **Escalations:** rate, guarantee and program drift; FY2020 over-predicted (COVID relief).



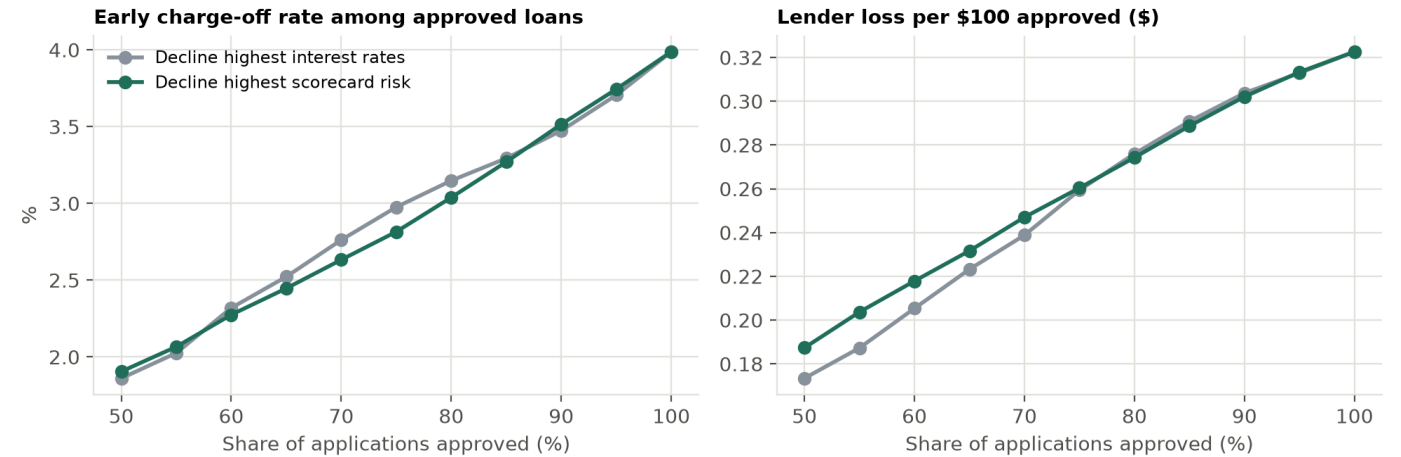
AUC on each held-out cohort. The dashed line is the exploratory spread-over-prime benchmark.

HELD-OUT RESULTS (FY2015–FY2020 COHORTS)

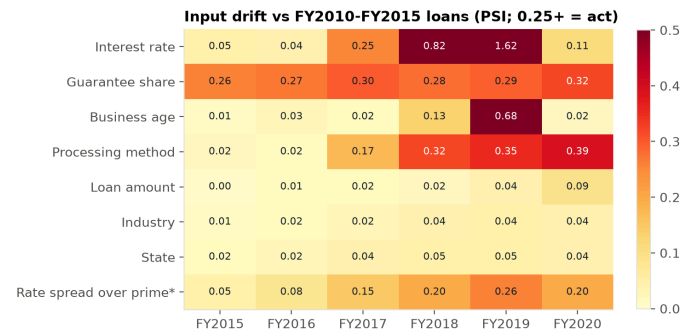
Model	AUC	95% CI	KS	Calibration slope
Interest rate alone	0.632	0.628–0.637	0.23	0.94
Rate spread over prime alone (exploratory)	0.674	—	—	—
Gradient boosting	0.677	0.672–0.681	0.26	0.89
Scorecard	0.684	0.680–0.689	0.28	1.03

Loan bootstrap within cohort (1,000 resamples). The scorecard beats boosting by 0.0075 (95% CI 0.004–0.011) and the spread benchmark by 0.010 (0.006–0.014).

Approval policies on held-out cohorts FY2015-FY2020



Approving the safest share of applicants. Lender loss = early charge-off amount times the unguaranteed share. The scorecard wins on default counts in the middle of the range; rate-based declines match or beat it in dollars.



LIMITATIONS

The outcome counts early charge-offs only (52–60% of charge-offs to date). There is no real-estate flag once term is removed, and no lender identity. Public data has no credit-bureau or financial-statement fields, so these results describe what origination data alone can do. The spread check came after seeing results and is labelled exploratory.

Data: U.S. Small Business Administration 7(a) FOIA data, snapshot as of June 30, 2026 (U.S. Government Works); prime rate from FRED (DPRIME). Not affiliated with or endorsed by the SBA.